SaaS Churn Prediction Project

Findings Report — Telco Customer Churn Dataset

Reproducibility Record:

- Dataset Name: Telco Customer Churn(IBM sample dataset)
- Version: Version 1
- URL: https://www.kaggle.com/datasets/blastchar/telco-customer-churn
- Download date: 10-09-2026

Data Record:

➔ Row Count: 7043
➔ Column Count: 21
➔ Data types: float64(1), int64(2), str(18).
➔ Churn rate: NO(5174), YES(1869)

Executive Summary

This project built and evaluated churn-prediction models on the Telco Customer Churn dataset (7,043 customers, 26% churn rate). Of the three model families compared, a class-weighted Logistic Regression performed best, catching 80% of customers who actually churned (ROC-AUC 0.845) while remaining the most interpretable of the models tested. The most valuable finding for the business is a clean three-tier risk segmentation: customers flagged "High Risk" churned at 55.8%, versus 4.6% for "Low Risk" customers a 12x difference that supports targeted, cost-effective retention spending rather than broad, undifferentiated campaigns.

Data & Methodology

The dataset contains 7,043 customers with 21 original fields covering demographics, account tenure, subscribed services, contract terms, billing, and churn status. Analysis followed a five-day structured process: exploratory audit, feature engineering, encoding, model comparison, and interpretability review, using an 80/20 stratified train-test split (random_state=42) to preserve the churn rate across both sets.

Data Cleaning Log

- TotalCharges: 11 rows contained blank strings instead of numeric values, all belonging to customers with tenure = 0 (new customers not yet billed). Converted to numeric and filled as 0 rather than dropped, preserving these customers in the analysis.
- Collapsed “No internet service” / “No phone service” into “No” across 7 service columns, since the InternetService field independently preserves that distinction.
- Correlation analysis showed avg_monthly_spend was a near-duplicate of TotalCharges (r=1.0), and TotalCharges itself was redundant with tenure and MonthlyCharges (r=0.8 and 0.7, respectively). Both avg_monthly_spend and TotalCharges were dropped from the final feature set to avoid multicollinearity.
- No duplicate customer IDs or logically invalid values (negative tenure or charges) were found.
- Remaining categorical fields were one-hot encoded (drop_first=True), producing a final feature matrix of 7,043 rows by 27 columns.
- Class-imbalance strategy: class weighting (class_weight='balanced') was chosen over synthetic oversampling (SMOTE), since it adjusts the model's loss function without fabricating synthetic customer records, keeping the decision boundary grounded in real observed data.

Model Comparison

Four models were trained and evaluated on the held-out test set (1,409 customers, 374 actual churners). Because the dataset is imbalanced (~26% churn), accuracy was deliberately excluded as a decision metric; a model that always predicts “no churn” would score roughly 74% accuracy while being useless. Recall, precision, F1, and ROC-AUC were used instead.

Model	Churn Recall	Churn Precision	Churn F1	ROC-AUC
Logistic Regression	0.80	0.51	0.62	0.845
Random Forest (default)	0.64	0.55	0.59	0.821
Random Forest (threshold 0.35)	0.80	0.49	0.61	0.821
XGBoost (default)	0.76	0.52	0.62	0.835

Logistic Regression was selected as the final model. It produced the highest ROC-AUC of any model tested, a threshold-independent measure of how well the model separates churners from non-churners, and matched or beat the tree-based models on F1 once recall was equalized. This is a notable finding: added model complexity (Random Forest, XGBoost) did not outperform a simple linear model on this dataset, most likely because the feature set, after cleaning, contains relatively few complex non-linear interactions for a more flexible model to exploit.

Key Churn Drivers

Coefficients from the final Logistic Regression model (fit on standardized features, so magnitudes are directly comparable) surfaced three leading drivers:

- Internet service type: customers with fiber optic internet show the single strongest churn risk in the model, a finding not visible in the initial exploratory plots (which focused on contract type, tenure, and payment method). Customers with no internet service at all show the opposite pattern, consistent with a simpler, lower-friction account relationship.
- Monthly charges: this feature carries a negative coefficient, meaning higher-paying customers appear less likely to churn in the model, but this is best read as an interaction effect with fiber service rather than a standalone causal signal. Fiber customers pay more per month and churn more; once the model accounts for fiber service separately, MonthlyCharges absorbs the remaining signal, which skews toward high-paying, non-fiber customers on stable, bundled plans. This should not be read as "raising prices reduces churn."
- Tenure and contract length: both confirm the exploratory findings: newer customers and those on flexible, month-to-month contracts churn at substantially higher rates than long-tenured customers on two-year contracts.

Additional Findings: Robustness & Error Analysis

Two further checks were run against the final Logistic Regression model to test how much a modeling choice mattered and where the model's predictions actually break down.

Does class weighting change the model, or just its trade-offs?

The final model was re-trained with no imbalance handling at all, holding the train/test split and feature set fixed, to isolate the effect of class_weight='balanced'.

Version	ROC-AUC	Churn Recall	Churn Precision
Weighted (final model)	0.845	0.80	0.51
Unweighted	0.8445	0.52	0.66

ROC-AUC is essentially unchanged between the two versions (0.845 vs. 0.8445), confirming that class weighting does not improve the model's underlying ability to rank customers by risk. It does change where the model draws its decision line: unweighted, it catches only 52% of churners but is more precise when it flags one (66%); weighted, it catches 80% of churners at the cost of more false alarms (51% precision). Class weighting is therefore best understood as a deliberate trade-off: the business is accepting more wasted retention outreach in exchange for fewer missed churners, not a genuine improvement in model quality.

Where the model fails: false negatives vs. false positives

Missed churners (false negatives, 74 customers) and false alarms (false positives, 293 customers) were compared on tenure, contract risk, and predicted probability to look for a pattern in the model's errors.

Group	Avg Tenure	% Month-to-Month	Avg Predicted Probability
False Negatives (missed churn)	31.4 months	50%	0.32
False Positives (false alarms)	21.5 months	91%	0.69

The model's errors are not random; they follow a clear, asymmetric pattern. Missed churners tend to look stable on paper: longer average tenure and an even mix of contract types, meaning they don't fit the model's strongest signal (new, month-to-month customers). These are the "quiet churners" the model is least equipped to catch. False alarms run the opposite way: they are overwhelmingly new, month-to-month customers (91%) with high average predicted probability (0.69, well above the 0.5 threshold); the model is confidently, not marginally, wrong about a meaningful share of loyal customers who happen to match the highest-risk profile on paper.

This is a useful limitation to act on: the model over-relies on the new-customer/month-to-month risk profile. A next iteration could benefit from features that distinguish stable long-term customers showing early warning signs from the broad pool of new customers who are flagged by profile alone but likely to stay.

Risk Segmentation & Business Recommendations

Customers in the test set were scored by predicted churn probability and split into three tiers using thresholds of 0.6 and 0.3:

Risk Tier	Customers	Actual Churn Rate	Suggested Action
High Risk	473 (33.6%)	55.8%	Proactive retention call + targeted discount or contract upgrade offer
Medium Risk	325 (23.1%)	25.2%	Low-cost automated nudge (email/SMS, loyalty perk, feature highlight)
Low Risk	611 (43.4%)	4.6%	No action — monitor only

The separation between tiers is strong: a High Risk customer is over 12 times more likely to churn than a Low Risk customer. This supports concentrating retention spend on the 33.6% of customers flagged High Risk, where the expected value of intervention is highest, while leaving the 43.4% flagged Low Risk untouched to avoid wasted spend on customers who were likely to stay regardless. The expected value of any intervention tier can be estimated as: (tier size) × (tier churn rate) × (customer lifetime value saved) − (tier size) × (intervention cost per customer), a calculation that requires lifetime-value and intervention-cost figures not present in this dataset, but that the business can apply directly using the tier sizes and churn rates above.

Limitations & Caveats

- The model's predicted probabilities run higher than the actual observed churn rate within each tier (e.g., High Risk customers average a 77.8% predicted probability against a 55.8% actual rate). This is a known side effect of class weighting: it shifts probabilities upward to compensate for class imbalance. Tier ranking remains valid, but the raw probabilities should not be read as literal likelihoods without a calibration step.
- This dataset is a single point-in-time snapshot, not a longitudinal record. The model's accuracy on customers observed at a different point in time, or under different market conditions, has not been tested.
- The model's errors are asymmetric rather than random; it under-catches longer-tenured churners and over-flags new, month-to-month customers who ultimately stay (see Additional Findings). Retention teams acting on High Risk flags should expect a meaningful share of false alarms concentrated in the newest customer segment.
- MonthlyCharges and fiber-optic service are correlated; their individual coefficients should be interpreted together, not as fully independent effects (see Key Churn Drivers).